

AWS Production Architecture & DevOps Proposal

Intelligent Technologies Ltd. — Technical Review Document

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1. Architecture Overview

Figure 1. AWS Production Architecture

The proposed architecture is designed to provide a secure, scalable, and cost-effective production environment. User requests are routed through Route 53, protected by AWS WAF and distributed using CloudFront when required. Traffic reaches the Application Load Balancer, which forwards requests to Amazon ECS Fargate tasks running across multiple Availability Zones for high availability. Amazon ECR stores Docker images, while CloudWatch provides monitoring and logging.

[Insert aws-architecture.png here]

2. CI/CD Pipeline

Figure 2. CI/CD Pipeline

Developers push source code to GitHub. GitHub Actions automatically builds the application, creates a Docker image, performs a Trivy security scan, pushes the image to Amazon ECR, and deploys the new version to Amazon ECS. Health checks validate the deployment before it is considered successful. GitHub Actions was selected for this pipeline because the development team already uses GitHub, it integrates directly with the repository, requires no dedicated CI server, and is more cost-effective than maintaining Jenkins for this project.

[Insert cicd.png here]

3. Deployment and Rollback Workflow

Figure 3. Deployment and Rollback Workflow

During deployment, a new ECS task is started while the existing version remains available. Health checks verify the new version. If the deployment succeeds, the Application Load Balancer routes traffic to the new task. If the health checks fail, traffic continues to the previous version and the failed deployment is automatically rolled back.

[Insert deployment.png here]

4. AWS Services

Service	Purpose
Route 53	DNS Management
WAF	Protect the application from common web attacks
CloudFront	Content delivery and caching
Application Load Balancer	Distribute incoming traffic
ECS Fargate	Run Docker containers

Service	Purpose
Amazon ECR	Store Docker images
CloudWatch	Monitoring and Logs
IAM	Access Management

5. DevOps Tools

Tool	Purpose
GitHub	Source Code Management
GitHub Actions	CI/CD Automation
Docker	Containerization
Trivy	Container Security Scanning
Terraform	Infrastructure as Code